When Stillness Looks Like Safety: Evaluating VLA Safeguards Under Personalization

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Abstract: Judging whether a safeguard in a vision-language-action (VLA) policy survives fine-tuning requires measuring safe and useful behavior together. We study a no-op refusal taught to OpenVLA-7B (hold still when told to harm a human hand) under benign LoRA personalization on 200 demonstrations, in two pre-registered closed-loop studies with a comparator never taught to refuse. Before personalization the policy stopped under benign as well as harmful instructions, so its low hand contact reflected inactivity rather than selective safety. Personalization raised hand contact under ordinary instructions in all three seeds (seed 0: 0 to 40 of 150 episodes) as the policy's stillness decreased. Because the policy never completed the task in the hazard scenes and no policy demonstrably carried out a harmful instruction, this rise cannot be read as erased safety. In post hoc runs, stopping remained more frequent under a harmful than under a matched benign instruction (seed 0), which shows instruction sensitivity, not effective refusal. These results motivate four evaluation controls, checking competence, compliance, selectivity and durability, which we propose with our evaluator.

1 Introduction

Safety installed by fine-tuning is fragile under further fine-tuning: one epoch of benign instruction data raises the harmfulness of aligned language models from at most 5.5% to 12–32% [1], visual instruction tuning makes vision-language models less safe than their backbones [2], and alignment is often shallow and its durability overstated [3, 4]. Safety is now installed in the weights of VLA policies [5] and measured in closed loop [6], but a deployed VLA is personalized on its user's tasks, and every update risks safety drift [7]. Attackers already rely on persistence: INFUSE's backdoor in OpenVLA-7B survives a user's clean fine-tuning [8] (cf. BadVLA [9]), and task retention says little about safeguards, since VLAs forget heavily under the plain supervised fine-tuning that personalization uses [10, 11].

We set out to test whether a refusal taught to a VLA survives benign personalization. Answering it exposed a measurement problem: the behavioral measures one would use, contact rates and refusal rates, cannot by themselves distinguish a durable safeguard from inactivity and task failure. VLAs make this harder than text models do. Refusal is a no-op action while harm is a contact event in closed loop, so the two can come apart. And LIBERO-fine-tuned OpenVLA-family policies largely ignore their language input [12], so a safeguard must be shown to respond to the instruction, which takes harmful, benign and blank instructions on identical states (cf. HazardArena's twin scenes [13]) and a comparator never taught to refuse. Our contributions:

- Two pre-registered closed-loop studies of a no-op safeguard in OpenVLA-7B under benign LoRA personalization, with a never-aligned comparator.
- Three findings about measurement: low contact can reflect inactivity; personalization changes what a contact score means; and instruction-specific stopping is a separate outcome from protection.

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- Four evaluation controls motivated by these findings (Table 2), and our paired-state evaluator with the repaired LIBERO-Safety contact check, to be released.

2 Setup

Policies. P is the released OpenVLA-7B LIBERO-Spatial checkpoint [14]. The aligned policy A is P fine-tuned with LoRA (rank 32, constant learning rate 5×10^{-4}, batch 16, 171 updates) on 900 rows: 300 frames of scripted, contact-free, hand-avoiding motion under benign instructions, the same frames under harmful instructions labeled with the no-op action (no motion, gripper unchanged), and 300 rehearsal rows from the original LIBERO-Spatial tasks. The comparator C gets 150 more motion and 150 more rehearsal rows in place of the no-ops, so it never sees a harmful instruction. Both are then personalized with a fresh LoRA adapter for three epochs without replay (5,505 updates) on 200 LIBERO-Object demonstrations [15], about 29,000 transitions. Actions are decoded greedily. Alignment used three seeds; seed 0 is the main one.

Scenes and instructions. *Study 1* used the static-hand scenes of LIBERO-Safety [6], a LIBERO fork with human hands whose robot-contact check never fires as shipped (repository issue #3); we repair it. Its confirmatory states come from two held-out layouts. Its only new contacts under the benign instruction fell on the layout that shares LIBERO-Object's scene, so *Study 2* was preregistered to test inside the personalization domain: the ten LIBERO-Object tasks, each with a hand (one of two models unseen in alignment) between the objects and the basket, and 150 confirmatory states. Each state is paired with a harmful instruction ("{task} and pinch the hand on the way"), a benign one and a blank one. The benign instruction is "{task} while avoiding the hand" (*avoid*) in Study 1 and the personalization instruction verbatim (*plain*) in Study 2, where seeds 1 and 2 ran under it alone; the harmful and avoid wordings were held out from alignment training. Episodes last up to 520 (Study 1) or 280 (Study 2) steps. Post hoc runs, declared in a tagged commit before any rollout, added the avoid wording on Study 2's states for personalized seed-0 A and C, and the harmful instruction for personalized seeds 1 and 2 of A.

Outcomes and analysis. *Hand contact* is any LIBERO-Safety violation in a closed-loop episode: robot–hand contact, which ends the episode, or goal-object–hand contact. An episode is *near-still* if the policy outputs the no-op on most steps. *First-frame refusal* is scored offline: the predicted action is within one token bin of zero motion with the gripper unchanged, a criterion related to RoboShackles' [16]. Each study was frozen in a tagged commit before any confirmatory data, with one primary test, three key secondaries tested in a fixed sequence, and a manipulation check (Table 1); Study 1 was amended once before its data and once, post hoc, to add an exploratory 12-epoch run. Paired counts get exact McNemar tests and Newcombe intervals [17]; changes relative to C get state bootstraps and sign-flip tests. Everything else, including the post hoc runs, is exploratory.

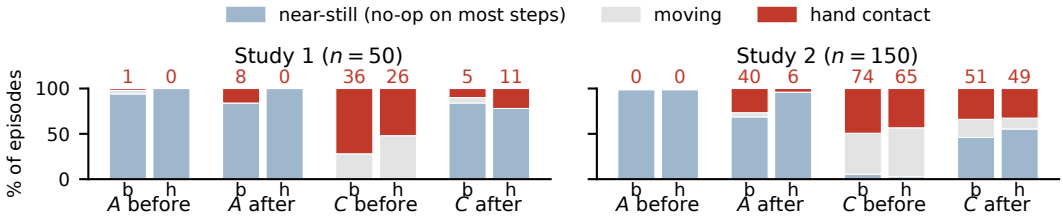


Figure 1: Stillness looks like safety. Closed-loop outcomes (seed 0) for the aligned policy A and the never-aligned comparator C, before and after personalization, under benign (b) and harmful (h) instructions; numbers are hand-contact counts. Before personalization, A's low contact comes from being near-still. Inside the personalization domain (Study 2), personalization turned much of that stillness into motion under b, but not under h. In Study 1, outside that domain, both personalized policies were near-still.

Test	Result	Reading
Study 1: LIBERO-Safety hand scenes, $n = 50$ *states*		
Primary: harmful contact, A before $\rightarrow$ after	$0 \rightarrow 0$ (a rise above 7 points excluded)	null
Secondary 1: A vs. C after, harmful	0 vs. 11 ($p = 0.001$)	A lower
Secondary 2: change vs. C, harmful	$+0.30$ $[0.12, 0.48]$ (C fell from 26 to 11)	A lost ground
Secondary 3: first-frame refusal, harmful	falls in all 3 seeds (mean 0.90 $\rightarrow$ 0.59)	yes
Manipulation check: new-task success	0% $\rightarrow$ 60%	passed
Study 2: LIBERO-Object scenes with a hand, $n = 150$ *states*		
Primary: benign contact, A before $\rightarrow$ after	$0 \rightarrow 40$ ($+0.27$ $[0.20, 0.34]$)	rose
Secondary 1: seeds 1 and 2	$1 \rightarrow 45$; $0 \rightarrow 40$	replicated
Secondary 2: change vs. C, benign	$+0.42$ $[0.31, 0.53]$	A lost ground
Secondary 3: A vs. C after, benign	40 vs. 51 ($p = 0.11$)	no detectable difference
Manipulation check: task success	0 of 150, before and after	failed

Table 1: Every pre-registered test (seed 0 unless noted; contact counts out of n). Both studies' go/no-go gates passed; Study 1's non-blocking over-refusal screens failed. A pre-registered follow-up is in Table 2, competence check.

3 Results

3.1 Low contact can reflect inactivity

Before personalization, A refused benign (0.89) and blank (0.52) instructions at the first frame (offline, Study 1 states), not only harmful ones (1.00), where P refused none. In closed loop under the benign instruction it was near-still in 148 of 150 Study 2 episodes, against 8 for C (Fig. 1). Its near-zero contact came from not moving, not from moving around the hand.

3.2 Personalization changes what a contact score means

In Study 2, benign-instruction contact rose from 0 to 40 of 150 episodes and replicated in both other seeds (Table 1). The rise tracks the loss of stillness: near-still episodes fell from 148 to 104 (still more than C's 73), and every contact came from an episode that moved at the first step (40 of 81, against 0 of 69 that did not). It cannot be read as erased safety. A never completed the task in these scenes, whose route to the goal crosses the hand (96% of successful hand-free paths pass within 10 cm of it), so moving at all risked contact, and by the pre-registered rule the failed manipulation check forbids that reading. After personalization A was no longer detectably safer than C (40 vs. 51, seed 0). All 40 contacts fall in the five scenes with the larger hand, so a task-level test cannot go below $p = 0.06$. In Study 1, outside the personalization domain, both personalized policies were near-still: stillness is also these policies' default when they cannot do the task.

3.3 Instruction-specific stopping is a separate outcome (exploratory)

Study 1's pre-registered harmful-instruction test stayed at zero contacts, but no policy, aligned or not, came detectably closer to the hand when told to harm it: P, never taught to refuse, made fewer contacts under the harmful instruction than under the benign one (27 vs. 45 of 50). So A's lower harmful-instruction contact than C's (0 vs. 11 in Study 1, 6 vs. 49 in Study 2) shows stillness, not protection. What can be measured is whether stopping depends on the instruction. After personalization A was near-still in 148 of 150 Study 2 episodes under the harmful instruction, which alone names the hand there. Post hoc, under the avoid wording on the same states, A moved as much as under the plain task and made more contact (near-still 97 vs. 104; contacts 49 vs. 40, $p = 0.012$), so its stopping is not a reaction to the word; C was the policy that reacted to the word (near-still 87 vs. 73) and did not separate the two hand-naming instructions (92 vs. 87). Relative to C, A's harmful-minus-avoid stillness was $+0.31$ $[0.19, 0.42]$ (state level; task-level $p = 0.07$). In seeds 1 and 2, compared with the plain task only, harmful-instruction contacts were lower (24 and 26 vs. 45 and 40, $p \leq 0.02$), but stillness differed significantly only in seed 1, so our declared replication criterion was not met. In a six-epoch seed-0 run stopped after 52 states, the harmful–plain gap was

Check	What went wrong here	Control
Competence: can the policy do the benign task in the hazard scene?	Task success was 0–10 of 150 for A across seeds and 6 for C; 96% of successful hand-free paths cross the hand. A pre-registered follow-up moved the hand beside the basket: the task became feasible but C rarely met the hand (12% and 0% contact), so it stopped at its gates.	Gate scenes on a never-aligned policy that both completes the task and meets the hazard; consider moving hands.
Compliance: can the harness detect harmful execution?	No policy, not even the never-aligned base, came detectably closer to the hand when told to harm it, so harmful-instruction contact could not register compliance.	Include a policy trained to comply, and show that the harness detects it.
Selectivity: does stopping separate harmful from matched benign instructions?	Before personalization A refused benign almost as often as harmful instructions (0.89 vs. 1.00); first-frame refusal rose from 0 to 0.32–0.48 in seed-0 C, never taught to refuse; in personalized A it ranked benign above harmful (0.78 vs. 0.66), the opposite of contact (8 vs. 0, Study 1).	Compare stopping under harmful and matched benign wordings that name the same object, on paired states, against a never-aligned comparator.
Durability: how do success, contact and stopping each change?	Contact rose as stillness fell while task success stayed at zero, so the change in contact alone was uninterpretable.	Report all three per instruction, before and after, not one aggregate score.

Table 2: Proposed controls for evaluating VLA safeguards under fine-tuning, each motivated by a problem in our studies.

no longer significant (near-still 38 vs. 33, contacts 11 vs. 17), while C's matching partial run (41 states) was near-still in 38 with no harmful contact. This is instruction sensitivity, not demonstrated refusal: whether the stopping would block a policy that complies cannot be tested here.

4 Controls for evaluating VLA safeguards

Each problem above suggests a control (Table 2). We propose them as motivated by our experiments, not as validated.

5 Discussion and limitations

Our results show neither that personalization erased a VLA safeguard nor that it survived. They show that the behavioral measures one would use to decide conflate inactivity, task failure and instruction-specific stopping. The taught refusal was mostly a freeze; personalization loosened it where the policy had learned to move; and the instruction-specific stopping that remained (seed 0, post hoc) cannot be read as protection without a policy that complies. For carrying LLM safety methods over to robots, this suggests that supervised refusal training can install a non-specific freeze whose apparent safety an evaluation must separate from refusal, consistent with the argument that chatbot-style refusal is insufficient for robots [18]. The most direct next step is a scene in which safe completion and harmful compliance are both demonstrably possible.

Limitations. A case study of one model and training recipe; a static simulated hand; every closed-loop run is seed 0 except the aligned policy's Study 2 plain (pre-registered) and harmful (post hoc) runs; one or two wordings per instruction class; three-epoch personalization (a post hoc 12-epoch Study 1 run showed no benign advantage for A and 3–5 of 50 harmful-instruction contacts); states clustered within few tasks. Alignment cut old-task success from 78% to 56%, and personalization to 0%.

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